SLR 1: Questions 1–3

1 Protocol basis and scope of this report

The protocol research questions are:

1. RQ1. How can nanosatellite missions and EPS be modeled in a general manner with minimal knowledge about their behavior?
2. RQ2. What are the claimed outcomes related to reliability, reproducibility, and integrability derived from EPS modeling in nanosatellite missions?
3. RQ3. How have verification procedures been integrated into EPS models prior to physical development?

The protocol extraction fields used as the organizing frame are: Bibliographic metadata, Modeling scope (system-level, EPS-level, requirements, architecture, budget, verification, etc.), MBSE language / method / tool, Research question(s) addressed, Claimed outcomes, Verification / validation linkage, Key findings, Limitations / gaps. The corpus summarized here contains 40 accessible extracted studies (21 core and 19 supporting), from 2013–2025. The four inaccessible kept corpus records listed in `pdf-status.md` are not included in the extraction synthesis.

2 Search, screening, retrieval, and extraction flow

Figure 1 gives the PRISMA-style flow reconstructed from the maintained search, screening, retrieval, and extraction artifacts.

The regenerated flow is: 199 raw merged search records, 118 duplicates removed, 81 unique records screened, 37 excluded during screening, 44 reports sought for retrieval, 4 not retrieved/inaccessible, and 40 extracted.

3 RQ relevance and modeling focus

Figure 4 summarizes the extraction-coded relevance to each protocol RQ. The key pattern is that RQ2 is coded as partial for all extracted studies, reflecting that claimed outcomes are usually MBSE/process claims rather than direct measured EPS reliability, reproducibility, or integrability outcomes. RQ3 has the smallest "yes" subset.

The focus categories answer the user's requested distinctions directly from extraction categories:

- Whole nanosatellite with EPS/power as a subsystem: 32 studies.
- Whole nanosatellite/generic with no explicit EPS: 7 studies (paper-0010, paper-0002, paper-0034, paper-0024, paper-0068, paper-0071, paper-0063).
- EPS-only or EPS-first model: 1 study (paper-0026).

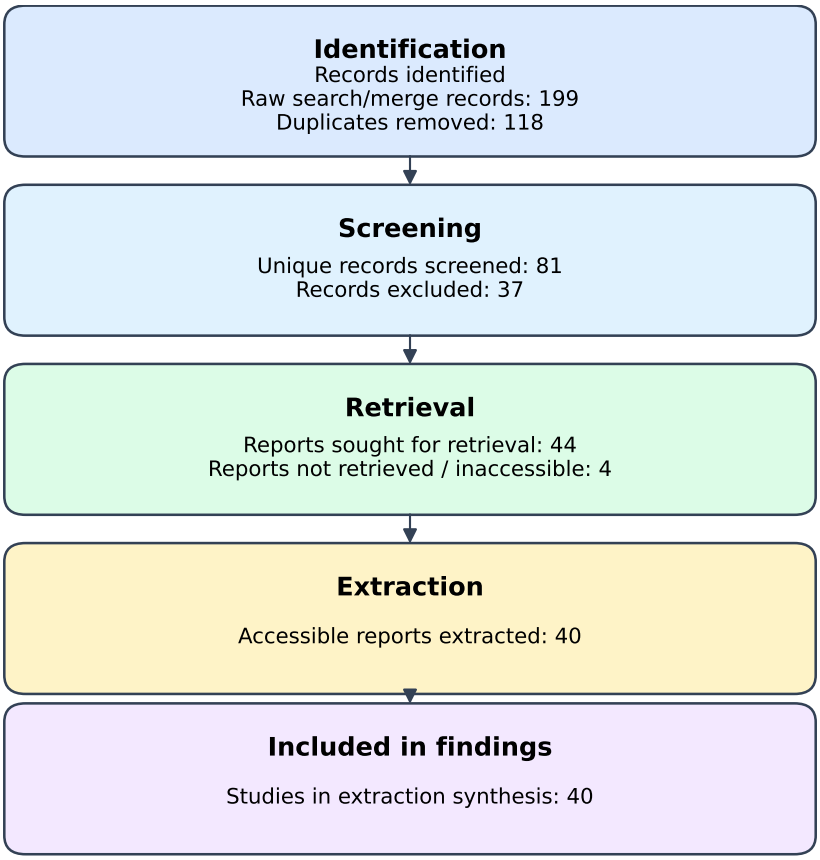


Figure 1: PRISMA-style flow for SLR1 extraction synthesis.

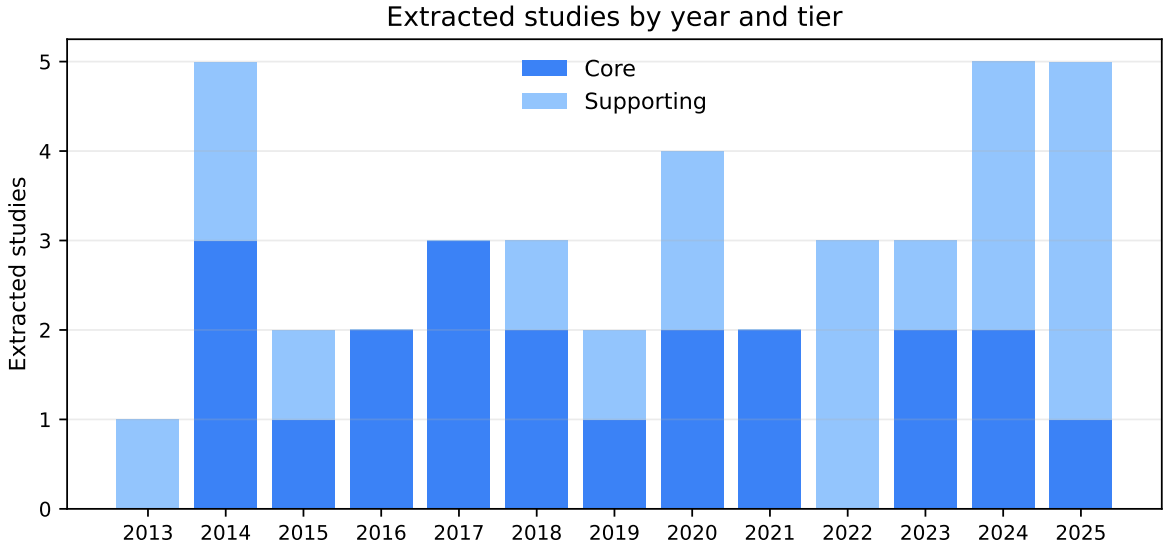


Figure 2: Extracted studies by publication year and screening tier.

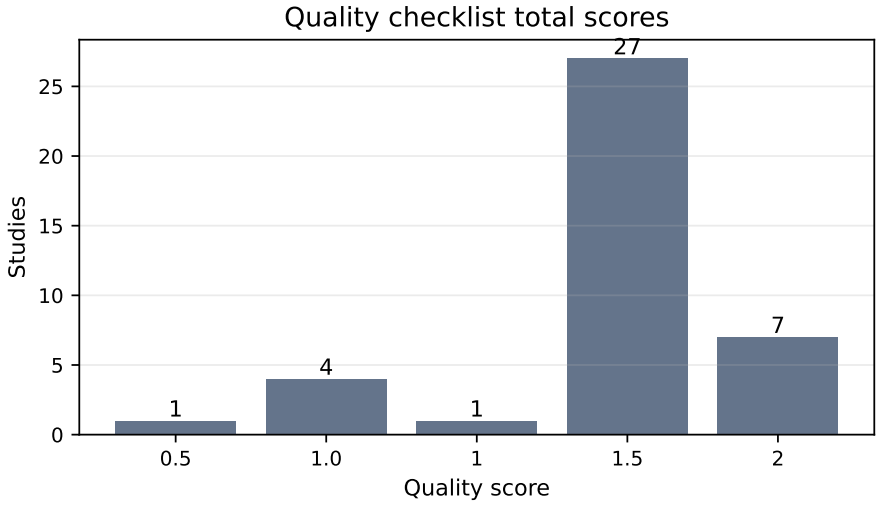


Figure 3: Quality checklist total scores recorded during extraction.

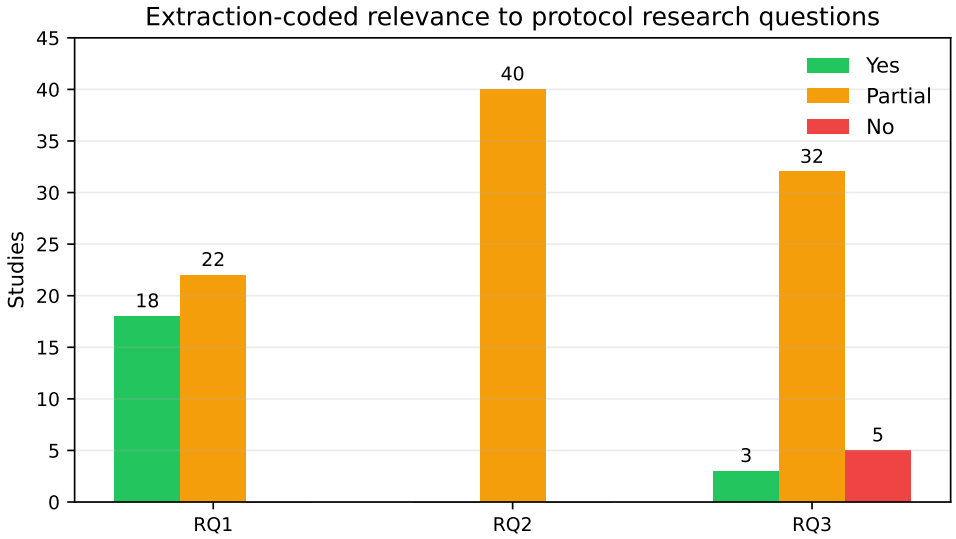


Figure 4: Extraction-coded relevance to each protocol research question.

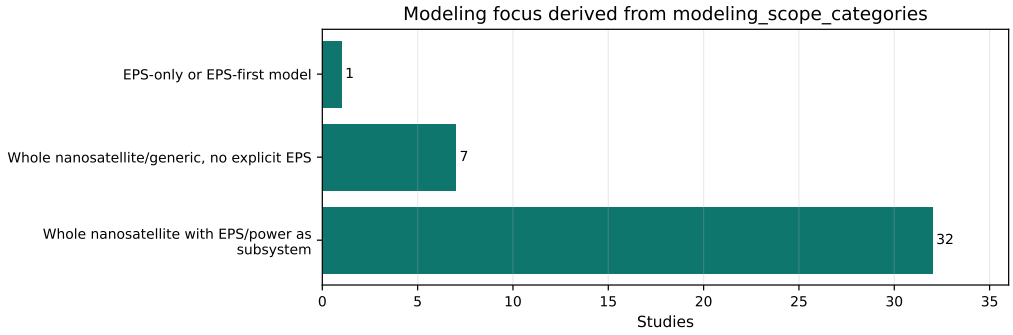


Figure 5: Modeling focus derived deterministically from `modeling_scope_categories`.



Figure 6: Study-level matrix showing modeling focus membership and RQ relevance.

3.1 Which studies fall in each modeling-focus group

Focus	Count	Studies
Whole nanosatellite with EPS/power as subsystem	32	paper-0003 (2013); paper-0001 (2014); paper-0008 (2014); paper-0012 (2014); paper-0079 (2014); paper-0004 (2015); paper-0022 (2015); paper-0016 (2016); paper-0031 (2016); paper-0005 (2017); paper-0015 (2017); paper-0007 (2018); paper-0013 (2018); paper-0080 (2018); paper-0077 (2019); paper-0011 (2020); paper-0039 (2020); paper-0014 (2021); paper-0033 (2021); paper-0027 (2022); paper-0046 (2022); paper-0037 (2023); paper-0023 (2024); paper-0029 (2024); paper-0059 (2024); paper-0061 (2024); paper-0078 (2024); paper-0041 (2025); paper-0043 (2025); paper-0053 (2025); paper-0054 (2025); paper-0081 (2025)
Whole nanosatellite/generic, no explicit EPS	7	paper-0010 (2014); paper-0002 (2017); paper-0034 (2019); paper-0024 (2020); paper-0068 (2020); paper-0071 (2022); paper-0063 (2023)
EPS-only or EPS-first model	1	paper-0026 (2023)

3.2 Generic whole-nanosatellite studies that do not model EPS explicitly

Table 2 lists the generic/no-EPS studies and the extraction-based reason each is not counted as EPS modeling.

Table 2: Why generic system-level studies were not counted as explicit EPS modeling.

Study	Short title	Why it is generic/no-EPS
paper-0010	Small Satellite Systems Design Methodology: A Formal and Agile...	formal/agile small-satellite design process and formal-methods V&V, not a subsystem model
paper-0002	A Model-Based Systems Engineering (MBSE) Approach for Defining...	behavior/use-case and functional decomposition of a generic CubeSat mission, not subsystem power design
paper-0034	Requirements Modeling Language and Automated Testing for CubeSats	requirements modeling language and automated-testing/traceability method, not a power subsystem model
paper-0024	Analysis of MBSE/PLM Integration: From Conceptual Design to...	MBSE–PLM information-flow analysis from conceptual to detailed design, not EPS architecture
paper-0068	Digital Engineering Information Exchange Model for Space...	digital-engineering information-exchange/DSM–DMM mapping for mission architecture, not EPS
paper-0071	A Model-Based Approach to Risk Management for RVSAT-1: a...	SysML risk-management model for a student nanosatellite, not EPS design or verification

Study	Short title	Why it is generic/no-EPS
paper-0063	Development of a Nanosatellite System Modeling Architecture...	nanosatellite system-modeling architecture with a communications-subsystem V&V demonstrator, not EPS

4 Categories arising in extraction fields

4.1 Modeling-scope categories

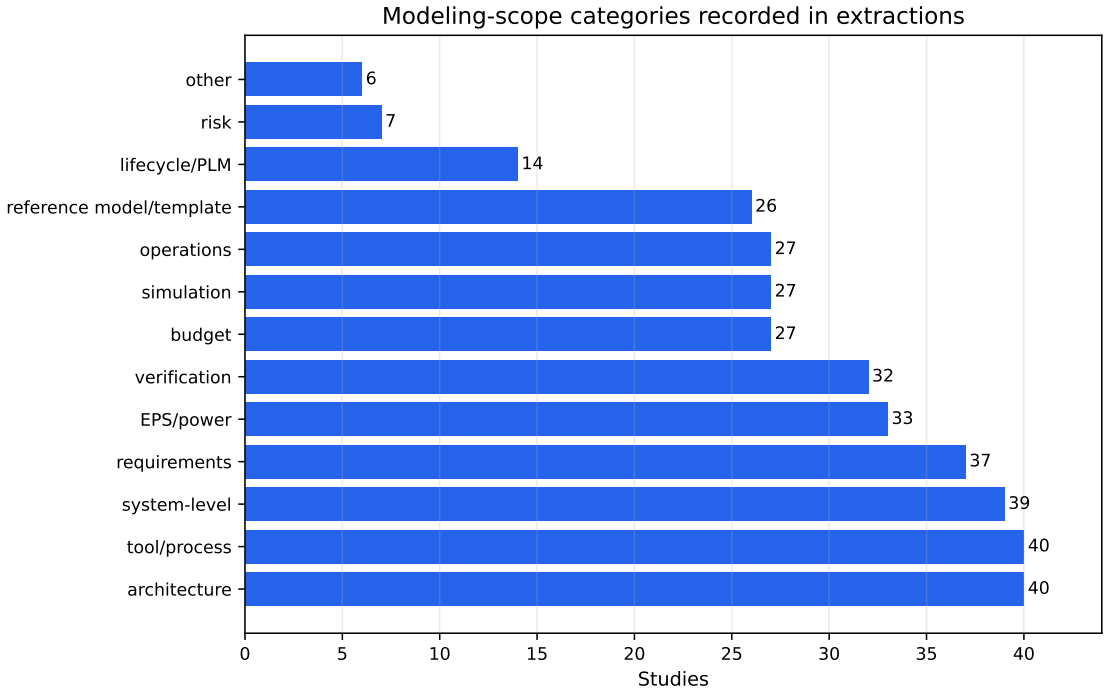


Figure 7: Multi-label modeling-scope categories recorded in the standardized extractions.

Category	n	Study IDs
architecture	40	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
tool/process	40	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081

Category	n	Study IDs
system-level	39	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
requirements	37	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
EPS/power	33	paper-0001, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081

Category	n	Study IDs
verification	32	paper-0001, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0031, paper-0033, paper-0034, paper-0037, paper-0043, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
budget	27	paper-0001, paper-0003, paper-0004, paper-0007, paper-0008, paper-0011, paper-0012, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0033, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0061, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
simulation	27	paper-0001, paper-0003, paper-0004, paper-0005, paper-0008, paper-0010, paper-0011, paper-0012, paper-0015, paper-0016, paper-0022, paper-0023, paper-0027, paper-0029, paper-0031, paper-0033, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0078, paper-0079, paper-0080
operations	27	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0008, paper-0011, paper-0012, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0031, paper-0033, paper-0037, paper-0043, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0077, paper-0078, paper-0080, paper-0081

Category	n	Study IDs
reference model/template	26	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0011, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0029, paper-0031, paper-0033, paper-0037, paper-0039, paper-0054, paper-0061, paper-0077, paper-0079, paper-0080, paper-0081
lifecycle/PLM	14	paper-0003, paper-0004, paper-0008, paper-0016, paper-0022, paper-0023, paper-0024, paper-0029, paper-0031, paper-0043, paper-0046, paper-0054, paper-0059, paper-0068
risk	7	paper-0007, paper-0008, paper-0010, paper-0054, paper-0061, paper-0071, paper-0078
other	6	paper-0014, paper-0024, paper-0037, paper-0039, paper-0068, paper-0071

4.2 MBSE languages, methods, and tools

The high-level extraction categories show SysML dominance but also a long tail of methods and toolchains. "Other" is present in all extraction files and is omitted from the main high-level tool figure; it remains in `report/data/category_counts.csv`.

Category	n	Study IDs
SysML	28	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0039, paper-0059, paper-0063, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080

Category	n	Study IDs
Cameo / MagicDraw / CATIA Magic	16	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0008, paper-0011, paper-0012, paper-0013, paper-0016, paper-0022, paper-0023, paper-0031, paper-0039, paper-0063, paper-0080
MATLAB / Simulink	12	paper-0001, paper-0003, paper-0004, paper-0008, paper-0011, paper-0012, paper-0013, paper-0023, paper-0063, paper-0078, paper-0079, paper-0080
STK / GMAT / DRAMA orbit tools	9	paper-0001, paper-0003, paper-0004, paper-0008, paper-0011, paper-0013, paper-0054, paper-0061, paper-0078
PLM / Teamcenter / Windchill / 3DExperience	9	paper-0004, paper-0008, paper-0016, paper-0022, paper-0024, paper-0031, paper-0046, paper-0059, paper-0068
OOSEM	7	paper-0004, paper-0015, paper-0016, paper-0022, paper-0031, paper-0063, paper-0080
ModelCenter / MBSE Analyzer	6	paper-0001, paper-0003, paper-0004, paper-0008, paper-0011, paper-0013
Excel / spreadsheets	5	paper-0003, paper-0061, paper-0063, paper-0080, paper-0081
ParaMagic / ParaSolver / acausal parametrics	5	paper-0003, paper-0004, paper-0008, paper-0079, paper-0080
OpenMBEE / XMI / model distribution	5	paper-0013, paper-0016, paper-0022, paper-0031, paper-0077
Capella / Arcadia	4	paper-0033, paper-0043, paper-0078, paper-0081
Nanospace / DSM / DMM / MB2DM	3	paper-0024, paper-0027, paper-0068
OPM / OPCloud / OPL	3	paper-0024, paper-0041, paper-0068
CPS abstraction	2	paper-0010, paper-0033
Papyrus / Eclipse	2	paper-0026, paper-0071
Docs-as-code / M30ML / YAML	2	paper-0037, paper-0081
Formal methods / PVS	1	paper-0010
RML / RMM	1	paper-0034
ForPlan / Julia / C2F	1	paper-0033
OPAL / SAPPhIRE	1	paper-0046
Gaphor / UML stereotypes	1	paper-0029

Category	n	Study IDs
Dymola / Modelica	1	paper-0059
Valispace	1	paper-0061
IDM-CIC / Simu-CIC / Opalis	1	paper-0053
System-as-Code / Python / JSON	1	paper-0081

4.3 Claimed outcomes

Outcome categories are extracted from the `claimed_outcomes` lists using deterministic keyword rules and with negated/absence statements separated. The most frequent positive outcome categories are: Efficiency, automation, or reduced time/cost/effort (32), Integration, interoperability, or data exchange (32), Communication, collaboration, or stakeholder alignment (30), Reuse, templates, standardization, or replicability (27), Traceability, consistency, or single source of truth (27).

Category	n	Study IDs
Efficiency, automation, or reduced time/cost/effort	32	paper-0002, paper-0003, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0078, paper-0079, paper-0080, paper-0081
Integration, interoperability, or data exchange	32	paper-0001, paper-0003, paper-0004, paper-0005, paper-0008, paper-0010, paper-0012, paper-0013, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0037, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081

Category	n	Study IDs
Communication, collaboration, or stakeholder alignment	30	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0027, paper-0031, paper-0037, paper-0039, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079
Reuse, templates, standardization, or replicability	27	paper-0001, paper-0003, paper-0004, paper-0005, paper-0008, paper-0011, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0039, paper-0041, paper-0043, paper-0053, paper-0054, paper-0063, paper-0071, paper-0077, paper-0079, paper-0080, paper-0081
Traceability, consistency, or single source of truth	27	paper-0001, paper-0002, paper-0004, paper-0005, paper-0007, paper-0013, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0034, paper-0037, paper-0041, paper-0043, paper-0046, paper-0054, paper-0059, paper-0063, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
Simulation, trade-space, budget, or early analysis support	25	paper-0001, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0012, paper-0013, paper-0016, paper-0022, paper-0026, paper-0029, paper-0031, paper-0033, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0079, paper-0080

Category	n	Study IDs
Reliability, mission success, quality, risk, or robustness	22	paper-0002, paper-0003, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0023, paper-0034, paper-0039, paper-0041, paper-0043, paper-0053, paper-0054, paper-0061, paper-0063, paper-0071, paper-0078, paper-0079, paper-0080
Verification, validation, test, or compliance support	20	paper-0001, paper-0003, paper-0004, paper-0008, paper-0010, paper-0013, paper-0015, paper-0023, paper-0033, paper-0034, paper-0039, paper-0043, paper-0053, paper-0059, paper-0061, paper-0063, paper-0071, paper-0077, paper-0079, paper-0080
Documentation, configuration, or change management	15	paper-0002, paper-0003, paper-0008, paper-0016, paper-0023, paper-0031, paper-0033, paper-0037, paper-0041, paper-0046, paper-0053, paper-0059, paper-0063, paper-0078, paper-0080
Openness, accessibility, or reproducibility	11	paper-0001, paper-0014, paper-0016, paper-0026, paper-0027, paper-0029, paper-0031, paper-0043, paper-0053, paper-0061, paper-0081

4.4 How verification/validation was linked to models

The most common verification-linkage categories are: V&V packages, reviews, plans, or procedures (28), Parametric, constraint, budget, or resource checks (20), Executable simulation linked to the model (16), Requirements linked through satisfy/verify/trace relations (13), General MBSE V&V statement only (8). These are multi-label categories; for example, a study can have requirements traceability, parametric checks, and external tool coupling at the same time.

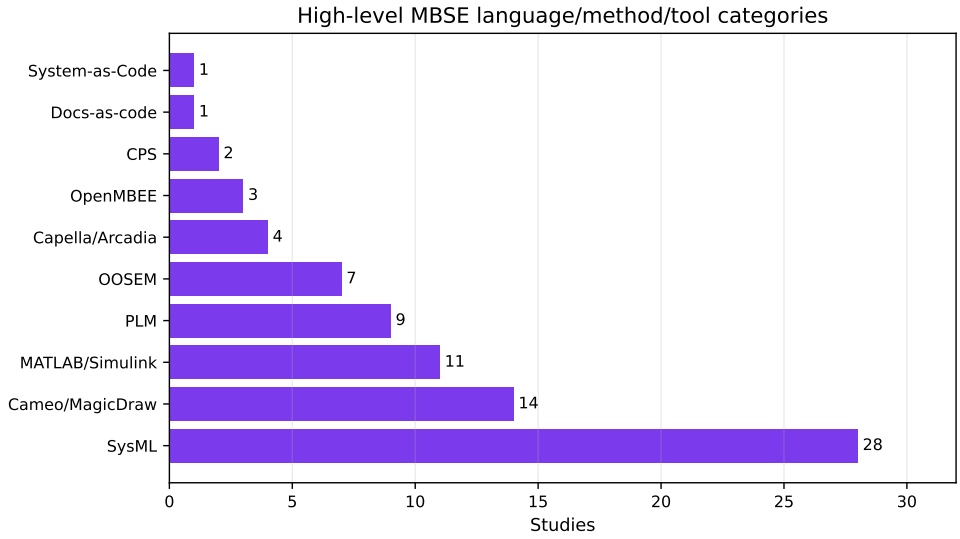


Figure 8: High-level MBSE language/method/tool categories, excluding the uninformative "Other" category.

Category	n	Study IDs
V&V packages, reviews, plans, or procedures	28	paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0013, paper-0014, paper-0015, paper-0016, paper-0023, paper-0024, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0053, paper-0059, paper-0061, paper-0071, paper-0077, paper-0078, paper-0080, paper-0081
Parametric, constraint, budget, or resource checks	20	paper-0001, paper-0003, paper-0008, paper-0023, paper-0026, paper-0027, paper-0029, paper-0033, paper-0039, paper-0041, paper-0043, paper-0053, paper-0059, paper-0061, paper-0063, paper-0068, paper-0077, paper-0078, paper-0079, paper-0080
Executable simulation linked to the model	16	paper-0001, paper-0003, paper-0005, paper-0010, paper-0011, paper-0027, paper-0029, paper-0033, paper-0041, paper-0043, paper-0053, paper-0059, paper-0077, paper-0078, paper-0079, paper-0080

Category	n	Study IDs
Requirements linked through satisfy/verify/trace relations	13	paper-0001, paper-0003, paper-0010, paper-0011, paper-0013, paper-0014, paper-0023, paper-0037, paper-0063, paper-0071, paper-0077, paper-0079, paper-0080
General MBSE V&V statement only	8	paper-0002, paper-0007, paper-0012, paper-0022, paper-0039, paper-0041, paper-0054, paper-0061
External analysis tool coupling or automatic feedback	7	paper-0001, paper-0003, paper-0059, paper-0063, paper-0078, paper-0079, paper-0080
Power/EPS resource requirement or feasibility check	6	paper-0001, paper-0008, paper-0043, paper-0053, paper-0059, paper-0077
No explicit model-linked V&V	6	paper-0002, paper-0012, paper-0024, paper-0029, paper-0046, paper-0068
Test cases, automated tests, or RTM/RMM mapping	5	paper-0008, paper-0023, paper-0034, paper-0078, paper-0080

4.5 Kinds of verification performed or proposed

Category	n	Study IDs
Requirement traceability or compliance checking	30	paper-0001, paper-0002, paper-0003, paper-0004, paper-0007, paper-0008, paper-0011, paper-0013, paper-0014, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0063, paper-0077, paper-0078, paper-0079, paper-0080
Simulation-based scenario/resource verification	18	paper-0001, paper-0003, paper-0005, paper-0008, paper-0010, paper-0015, paper-0023, paper-0027, paper-0029, paper-0033, paper-0034, paper-0041, paper-0043, paper-0053, paper-0054, paper-0059, paper-0078, paper-0079

Category	n	Study IDs
Parametric, budget, or margin verification	18	paper-0001, paper-0003, paper-0008, paper-0023, paper-0026, paper-0027, paper-0029, paper-0033, paper-0039, paper-0041, paper-0043, paper-0053, paper-0059, paper-0063, paper-0077, paper-0078, paper-0079, paper-0080
No concrete verification performed or only generic V&V claim	9	paper-0002, paper-0012, paper-0022, paper-0024, paper-0029, paper-0039, paper-0046, paper-0061, paper-0068
Review, inspection, validation package, or procedure planning	8	paper-0005, paper-0011, paper-0013, paper-0014, paper-0015, paper-0023, paper-0077, paper-0081
Test-case or automated-test verification	5	paper-0008, paper-0023, paper-0034, paper-0078, paper-0080
Model consistency or tool validation	3	paper-0010, paper-0043, paper-0081

5 Contributions and limitations

The contribution taxonomy shows what each paper primarily added to the corpus. The most frequent contribution categories are: Toolchain, data integration, or interoperability (40), Method, framework, process, or workflow (40), System-/mission-level nanosatellite model (39), Mission or case-study application (39), Simulation, trade study, executable analysis, or sizing (38).

Category	n	Study IDs
Toolchain, data integration, or interoperability	40	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081

Category	n	Study IDs
Method, framework, process, or workflow	40	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
System-/mission-level nanosatellite model	39	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081

Category	n	Study IDs
Mission or case-study application	39	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0068, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
Simulation, trade study, executable analysis, or sizing	38	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081

Category	n	Study IDs
Requirements, traceability, V&V, or testing linkage	38	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0010, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0063, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
EPS, power, resource-budget, or electrical-architecture representation	34	paper-0001, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0037, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0054, paper-0059, paper-0061, paper-0071, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
Reference model, template, or reusable architecture	30	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0007, paper-0008, paper-0011, paper-0012, paper-0013, paper-0014, paper-0015, paper-0016, paper-0022, paper-0023, paper-0026, paper-0029, paper-0031, paper-0033, paper-0037, paper-0039, paper-0053, paper-0054, paper-0061, paper-0068, paper-0071, paper-0077, paper-0079, paper-0080, paper-0081

Category	n	Study IDs
Risk or technical-measurement model	14	paper-0002, paper-0007, paper-0008, paper-0010, paper-0011, paper-0013, paper-0014, paper-0022, paper-0054, paper-0061, paper-0071, paper-0077, paper-0078, paper-0080

Limitations are also multi-label. The most frequent limitation categories are: Prototype, incomplete maturity, or future work (31), Limited scope, single case, or partial subsystem/mode coverage (24), No quantitative evaluation, weak validation, or no threats discussion (24), Data, model-quality, source, or assumption limitation (18), Tool dependency, interoperability, licensing, or learning curve (16). Several studies do not provide formal limitations/threats-to-validity discussions; those are explicitly separated rather than silently ignored.

Category	n	Study IDs
Prototype, incomplete maturity, or future work	31	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0008, paper-0011, paper-0013, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0031, paper-0033, paper-0034, paper-0037, paper-0041, paper-0043, paper-0046, paper-0054, paper-0061, paper-0063, paper-0068, paper-0077, paper-0078, paper-0079, paper-0080, paper-0081
Limited scope, single case, or partial subsystem/mode coverage	24	paper-0001, paper-0002, paper-0003, paper-0004, paper-0008, paper-0012, paper-0016, paper-0022, paper-0023, paper-0024, paper-0026, paper-0027, paper-0029, paper-0039, paper-0041, paper-0043, paper-0046, paper-0053, paper-0059, paper-0061, paper-0063, paper-0071, paper-0078, paper-0079

Category	n	Study IDs
No quantitative evaluation, weak validation, or no threats discussion	24	paper-0001, paper-0002, paper-0004, paper-0007, paper-0008, paper-0010, paper-0011, paper-0013, paper-0015, paper-0016, paper-0022, paper-0023, paper-0029, paper-0031, paper-0033, paper-0034, paper-0039, paper-0041, paper-0043, paper-0046, paper-0059, paper-0061, paper-0068, paper-0071
Data, model-quality, source, or assumption limitation	18	paper-0001, paper-0002, paper-0003, paper-0004, paper-0005, paper-0008, paper-0010, paper-0012, paper-0024, paper-0026, paper-0027, paper-0029, paper-0033, paper-0037, paper-0039, paper-0043, paper-0046, paper-0061
Tool dependency, interoperability, licensing, or learning curve	16	paper-0003, paper-0004, paper-0008, paper-0014, paper-0023, paper-0027, paper-0029, paper-0039, paper-0041, paper-0046, paper-0059, paper-0061, paper-0063, paper-0071, paper-0080, paper-0081
Organizational, adoption, or training constraint	14	paper-0003, paper-0005, paper-0008, paper-0011, paper-0012, paper-0015, paper-0022, paper-0037, paper-0039, paper-0046, paper-0071, paper-0077, paper-0078, paper-0080
Generalizability or mission-specific tailoring constraints	9	paper-0012, paper-0013, paper-0015, paper-0016, paper-0022, paper-0029, paper-0039, paper-0054, paper-0077
No limitations discussed	1	paper-0053

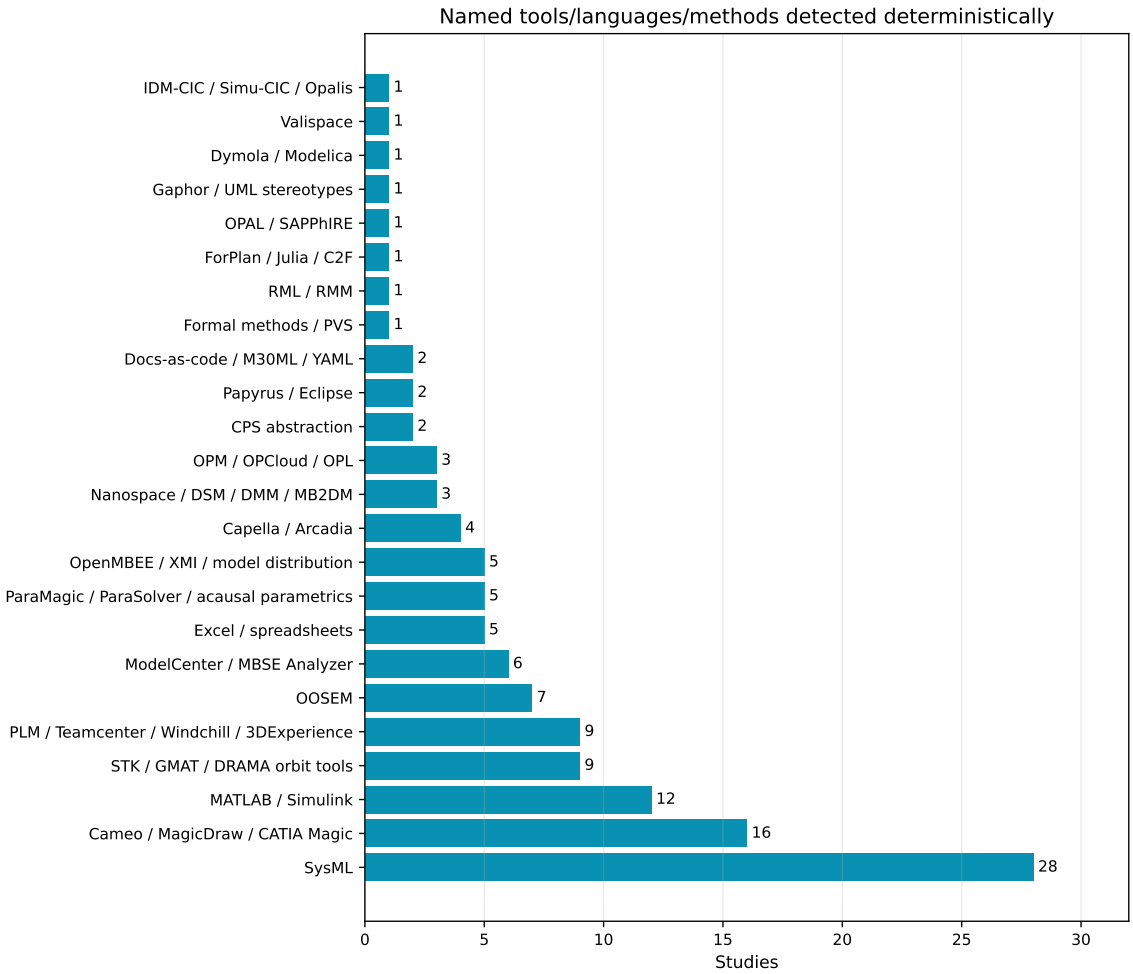


Figure 9: Named tools/languages/methods detected from `mbse_language_method_tool` using deterministic keyword rules.

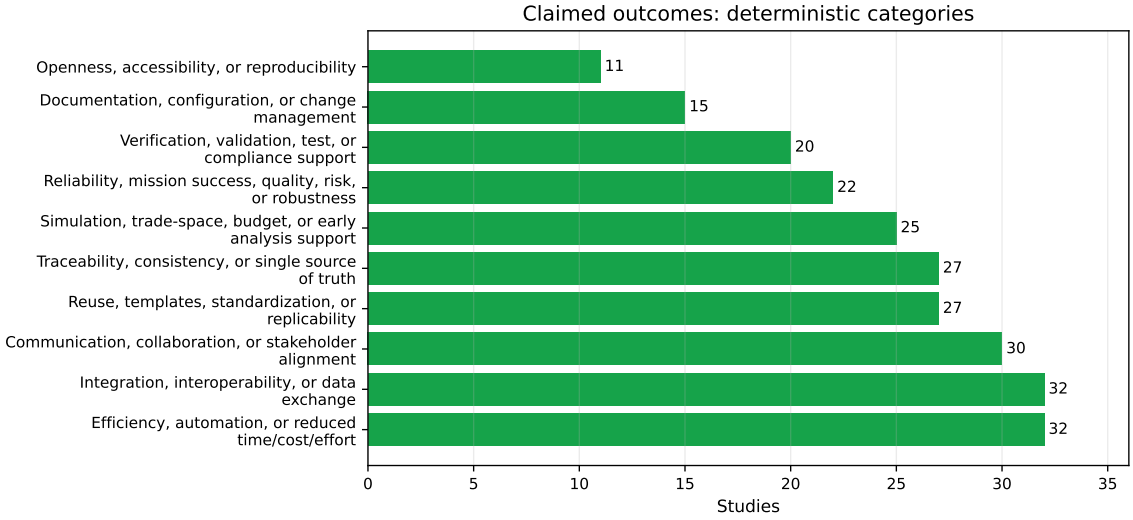


Figure 10: Positive claimed-outcome categories detected in the extraction field.

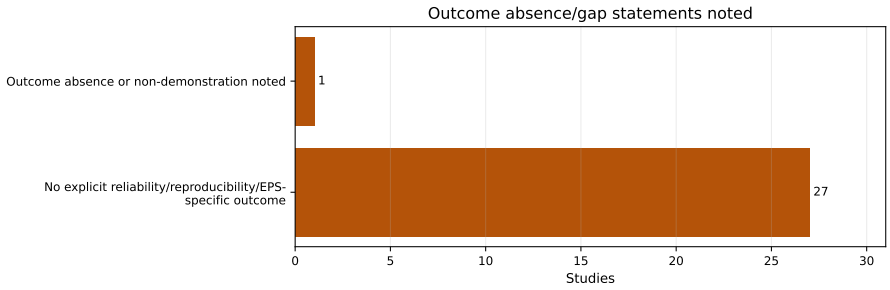


Figure 11: Outcome absence/gap categories explicitly noted in extraction bullets.

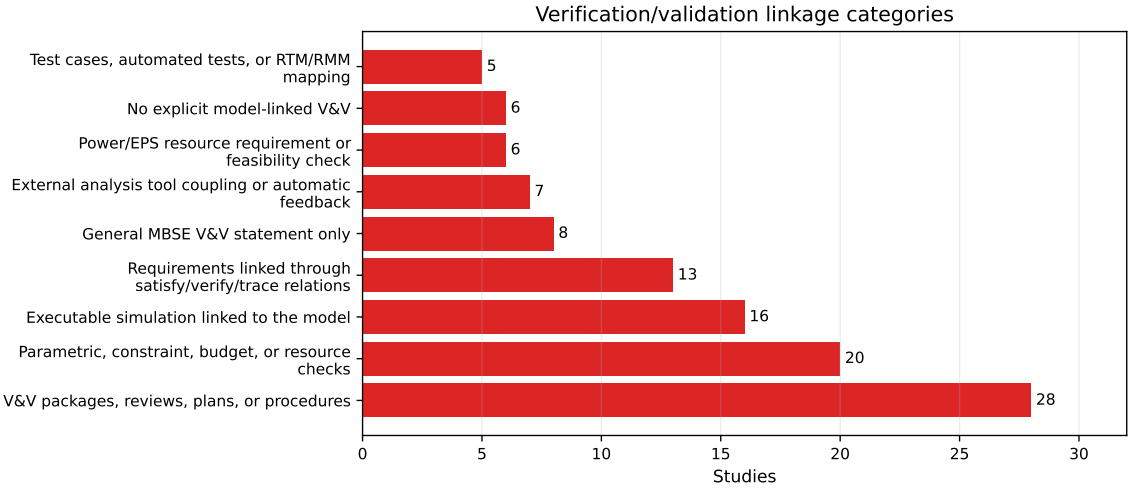


Figure 12: How verification/validation was linked to the model in the extractions.

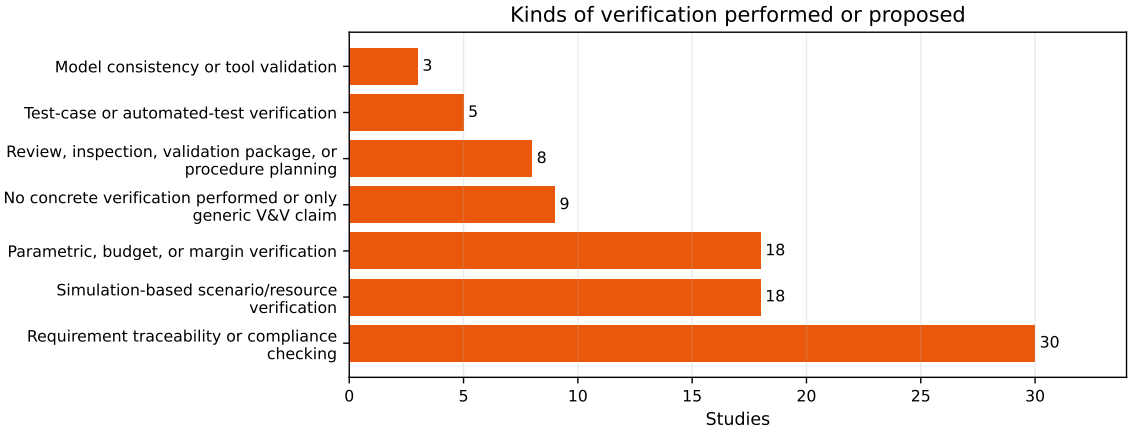


Figure 13: Kinds of verification performed or proposed, derived from V&V linkage and key-finding fields.

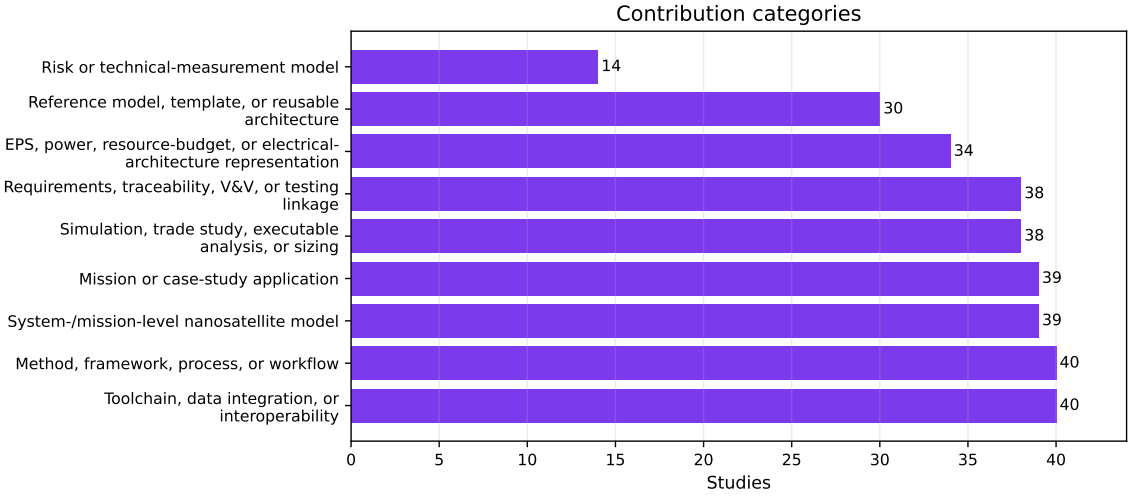


Figure 14: Contribution categories derived from key findings, modeling scope, and tool/method fields.

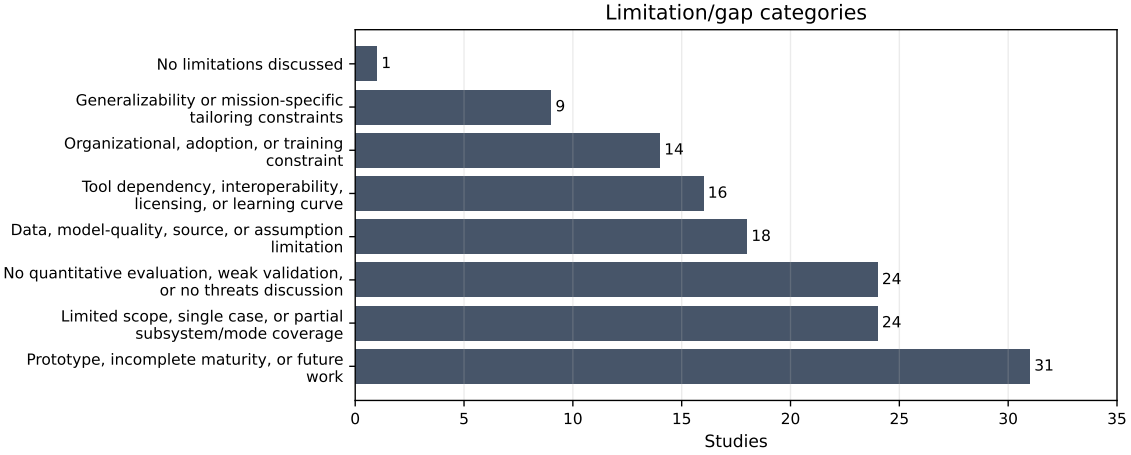


Figure 15: Limitation/gap categories derived from the standardized limitation fields.

6 Study-level contribution and limitation map

The following landscape table supports paper-by-paper inspection of what each extraction contributed and what limitations/gaps were recorded.

Study	Short title	Contribution categories	Limitation categories
paper-0003	Model Based Systems Engineering (MBSE) Applied to Radio...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0001	Integrated Model-Based Systems Engineering (MBSE)...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work
paper-0008	Enterprise modeling for CubeSats	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0010	Small Satellite Systems Design Methodology: A Formal...	Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; No quantitative evaluation, weak validation, or no threats discussion
paper-0012	Rapid model-based interdisciplinary design of a...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Generalizability or mission-specific tailoring constraints; Limited scope, single case, or partial subsystem/mode coverage; Organizational, adoption, or training constraint
paper-0079	Model-based systems engineering to support the...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; Prototype, incomplete maturity, or future work

Study	Short title	Contribution categories	Limitation categories
paper-0004	Developing a CubeSat Model-Based System Engineering...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0022	CubeSat Model-Based System Engineering (MBSE) Reference...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Generalizability or mission-specific tailoring constraints; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work
paper-0016	Developing a CubeSat Model-Based System Engineering...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Generalizability or mission-specific tailoring constraints; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work
paper-0031	CubeSat Model-Based System Engineering (MBSE) reference...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work
paper-0002	A Model-Based Systems Engineering (MBSE) Approach for...	Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work

Study	Short title	Contribution categories	Limitation categories
paper-0005	Developing a CubeSat Model-Based System Engineering...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work
paper-0015	Validation and Verification of MBSE-compliant CubeSat...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Generalizability or mission-specific tailoring constraints; No quantitative evaluation, weak validation, or no threats discussion; Organizational, adoption, or training constraint
paper-0007	A model-based systems engineering approach for...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	No quantitative evaluation, weak validation, or no threats discussion
paper-0013	Developing a CubeSat Model-Based System Engineering...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Generalizability or mission-specific tailoring constraints; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work
paper-0080	Application of Model-Based Systems Engineering in Small...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve

Study	Short title	Contribution categories	Limitation categories
paper-0034	Requirements Modeling Language and Automated Testing...	Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work
paper-0077	Developing a CubeSat System MBSE Reference Model –...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Generalizability or mission-specific tailoring constraints; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work
paper-0011	Development and Application of the CubeSat System...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	No quantitative evaluation, weak validation, or no threats discussion; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work
paper-0024	Analysis of MBSE/PLM Integration: From Conceptual...	Method, framework, process, or workflow; Mission or case-study application; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; Prototype, incomplete maturity, or future work
paper-0039	Applying a Model-Based Approach to Develop a...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Generalizability or mission-specific tailoring constraints; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Organizational, adoption, or training constraint; Tool dependency, interoperability, licensing, or learning curve
paper-0068	Digital Engineering Information Exchange Model for...	Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work

Study	Short title	Contribution categories	Limitation categories
paper-0014	Mission Engineering and the CubeSat System Reference Model	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Tool dependency, interoperability, licensing, or learning curve
paper-0033	Towards Simulation of CubeSat Operational Scenarios...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work
paper-0027	MBSE Challenges in the Concurrent Preliminary Design of...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0046	Investigating the Effectiveness of a Causal MBSE...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0071	A Model-Based Approach to Risk Management for RVSAT-1:...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Organizational, adoption, or training constraint; Tool dependency, interoperability, licensing, or learning curve
paper-0026	Open-source CubeSat MBSE Approach to Address...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; Prototype, incomplete maturity, or future work

Study	Short title	Contribution categories	Limitation categories
paper-0037	Model Based Systems Engineering with a Docs-as-Code...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work
paper-0063	Development of a Nanosatellite System Modeling...	Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0023	Effective Model-Based Systems Engineering framework for...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0029	Open-Source Data Formalization through Model-Based...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Generalizability or mission-specific tailoring constraints; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0059	An Implementation of Integrated Approach in Product...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Tool dependency, interoperability, licensing, or learning curve
paper-0061	Concurrent Model-Based Approach for CubeSat Mission Design	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve

Study	Short title	Contribution categories	Limitation categories
paper-0078	Preliminary Design of Satellite Systems through the...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; Organizational, adoption, or training constraint; Prototype, incomplete maturity, or future work
paper-0041	Applying Model-Based Systems Engineering on CubeSat...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve
paper-0043	Towards a Unified Modeling and Simulation Framework for...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Data, model-quality, source, or assumption limitation; Limited scope, single case, or partial subsystem/mode coverage; No quantitative evaluation, weak validation, or no threats discussion; Prototype, incomplete maturity, or future work
paper-0053	CONCURRENT ENGINEERING PROCESS FOR ELECTRICAL...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Limited scope, single case, or partial subsystem/mode coverage; No limitations discussed
paper-0054	Integrating Model-Based Systems Engineering into...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Risk or technical-measurement model; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Generalizability or mission-specific tailoring constraints; Prototype, incomplete maturity, or future work
paper-0081	Automating Object-Oriented Functional Modeling for...	EPS, power, resource-budget, or electrical-architecture representation; Method, framework, process, or workflow; Mission or case-study application; Reference model, template, or reusable architecture; Requirements, traceability, V&V, or testing linkage; Simulation, trade study, executable analysis, or sizing; System-/mission-level nanosatellite model; Toolchain, data integration, or interoperability	Prototype, incomplete maturity, or future work; Tool dependency, interoperability, licensing, or learning curve

7 Generated artifacts and reproducibility

This report is regenerated by running:

```
python3 SLR/SLR1/scripts/generate_extraction_report.py
```

The script writes the figures under `SLR/SLR1/report/figures/`, tabular derived data under `SLR/SLR1/report/dat`
and this LaTeX file at `SLR/SLR1/report/extraction_findings.tex`. The deterministic category
rules are persisted in the script.